

Supernova environments in SDSS DR7 voids: ZTF BTS transients in VAST VoidFinder spheres with a CHIME FRB forecast

Himel Das,¹★

¹Government Azizul Haque College, Bogura 5800, Bangladesh

Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

We cross-match VAST VoidFinder SDSS DR7 (1163 voids, $z < 0.114$, median $R_{\max} = 11.9 h^{-1}\text{Mpc} = 17.6\text{Mpc}$, median $R_{\text{eff}} = 15.5 h^{-1}\text{Mpc} = 22.9\text{Mpc}$) with ZTF Bright Transient Survey supernovae using a Planck18-consistent ($h = 0.6766$) KD-tree ball query with 1-deg SDSS mask (quantized; edge test in text). Of 6873 BTS transients at $z = 0.01\text{--}0.114$, 2465 lie in-mask: 147 void ($r/R_{\max} < 0.8$), 195 shell ($0.8\text{--}1.0$), 2080 wall (43 outside). Cosmo SNe (TDE/Ca-rich/other excluded): 2367 total, 142 void (118 SN Ia; subtype audit pending TNS). Volume-limited $z \leq 0.06$: 1270 SNe total, 69 void (46 Ia). Raw Poisson sensitivity to a void–wall Hubble offset is ~ 0.035 mag at 95 per cent including systematics floor; blinded SALT2 standardisation with mass, peculiar-velocity and selection controls is in progress. CHIME Catalog 1 (600 bursts, 524 sources) is used for footprint and error-budget demonstration only; unlocalised $z < 0.114$ stacking is underpowered by $> 10\times$ and joint constraints are deferred to Catalog 2. Catalog, randoms design and pipeline will be released.

Key words: large-scale structure of Universe – supernovae: general – fast radio bursts – methods: statistical

Table 1. Prior work vs this work.

Study	Voids	z	SNe	FRBs
Wang	lumped	< 0.05	hetero.	0 in voids
Tsaprazi	dens.+web	< 0.036	BTS+CLU	–
Aubert	DR7+Vor.	DR2 Ia	stretch	–
Sharma	BOSS VIDE	0.2–0.7	–	3455 stack
This work	VAST $R_{\max}$	< 0.114	BTS+slice	Cat1 fcst

1 INTRODUCTION

Cosmic voids fill most of the volume. Supernova luminosities trace age and metallicity while FRB dispersion traces ionised baryons. Table 1 summarises prior work: void-transient demographics (Wang et al. 2026; Tsaprazi et al. 2021), ZTF Ia environments (Aubert et al. 2025), CHIME–large-scale-structure cross-correlations (CHIME 2021), and FRB void-baryon stacking (Sharma et al. 2026). No published paper jointly uses VoidFinder DR7 spheres (Douglass et al. 2023) with flux-limited BTS (Perley et al. 2020; Fremling et al. 2019) and CHIME in the same low-redshift volumes; this work provides a supernova-main result with an FRB forecast in that geometry. Distances assume Planck Collab. (2020) Planck18 ($H_0 = 67.66\text{ km s}^{-1}\text{Mpc}^{-1}$, $\Omega_m = 0.3097$); void-catalog lengths in h^{-1} units use $h = 0.6766$ throughout.

2 DATA

VAST Zenodo 7406035 v1.3.0 (Douglass et al. 2023); BTS Explorer quality and purity cuts (2026-09-30, 7143 rows) (Perley et al. 2020; Fremling et al. 2019); CHIME Catalog 1 Vizier J/ApJS/257/59/table2 (600 bursts, 524 sources) (CHIME 2021) with Catalog 2 (CHIME 2026) for exposure; NASA-Sloan Atlas v1_0_1 downloading for parent masses; Planck Collab. (2020) throughout.

3 METHODS

Void centres recomputed from RA, Dec, D ($D\text{Mpc} = \text{Mpc}/h/0.6766$, Planck18 $H_0 = 67.66$) in one consistent frame; stored-XYZ residual median 0.00Mpc . $R_v \equiv R_{\max}$ core spheres (median $11.9 h^{-1}\text{Mpc} = 17.6\text{Mpc}$); R_{eff} as systematics branch. Membership via ball query, $r/R_{\max} = \min$, shell $0.8\text{--}1.0$ excluded from primary. Mask from NSA_main_mask.pickle (360,180 bool, 1-deg quantized). Edge test: eroding one cell removes 5.6% of mask cells, 108/2367 SNe (3 void), 12/1163 void centres; cosmo void fraction shifts $0.0744 \rightarrow 0.0766$ (3% relative, subdominant to Poisson 8%). Randoms: 200000 uniform comoving ($z = 0.01\text{--}0.114$) per seed (42,43,44), ~ 35200 in-mask each; cos-weighted sky fraction 0.177 (naive cell mean 0.141 is wrong for equirectangular grids); expected $f_{\text{void}} = 0.1896 \pm 0.0018$ (seed spread), $f_{\text{shell}} = 0.166$, $f_{\text{wall}} = 0.627$ finite ($f_{\text{out}} = 0.018$); $V_{\text{eff,tot}} = 8.80 \times 10^7\text{Mpc}^3$. Observed counts read from the parquet (no hand copies). Blinded SALT2 (Guy et al. 2007) hierarchical Δ_{HR} with Tripp standardisation (Tripp 1998), peculiar-velocity covariance, mass-step joint fit and BTS efficiency planned; this version reports mask-corrected counts with randoms volumes and split apparent rate ratios only.

★ E-mail: contact via ORCID 0009-0003-7698-1255

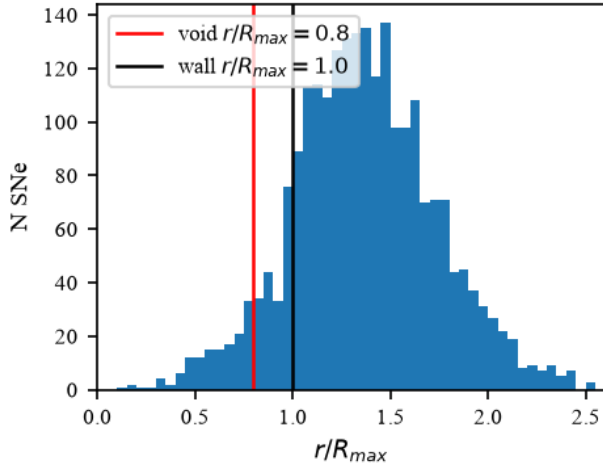


Figure 1. $r/R_{\max}$ for $N = 2367$ cosmology-grade SNe (void $r/R_{\max} < 0.8$, $N = 142$; wall > 1.0). Vertical lines mark the analysis thresholds.

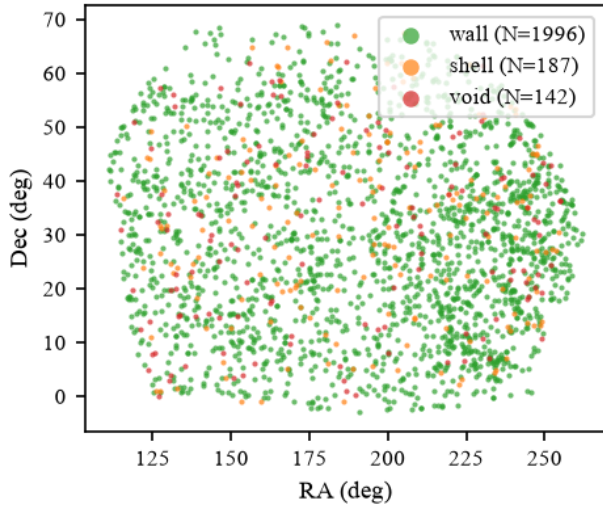


Figure 2. Masked RA–Dec for cosmology-grade SNe: wall ($N = 1996$, green), shell ($N = 187$, orange), void ($N = 142$, red). Outside-mask and edge-excluded objects omitted.

4 RESULTS

Cuts flow: 7143 $\rightarrow$ 6873 (z) $\rightarrow$ 2465 (mask). Cosmo SNe 2367: void 142, wall 1996 (187 shell, 42 outside; Table 2). Observed void fraction $142/2367 = 0.060$ versus randoms 0.190: apparent (uncorrected) void/wall ratios 0.24 all, 0.26 Ia-only (118/1493), 0.16 CC-only (24/503); Poisson+Monte Carlo rel. err 0.09–0.21. No efficiency or host-mass correction applied, so these are apparent deficits in a flux-limited sample, not intrinsic rates. Table 2 sums exactly; shell/outside rows are tabulated, not dropped. Void holds 118 SN Ia (subtype audit pending TNS). Volume slice $z \leq 0.06$: 1270 SNe total, 69 void (46 Ia). Threshold grid (SNe): 0.6:56, 0.7:88, 0.8:142, 0.9:220, 1.0:329, 1.1:531. Redshift stability $z_{\min}$ 0.01:142, 0.02:135, 0.025:130, 0.03:126 void. Void median z 0.064 versus wall 0.056 confirms Malmquist gradient. FRB error budget (Cat1, following

Macquart et al. (2020) and void-decrement simulations (Walker et al. 2024)): per-sightline $\sigma_{\text{tot}} \sim 150\text{--}200 \text{ pc cm}^{-3}$ (IGM 80–120, host 80–150, Milky Way residual 20–40 with NE2001–YMW16 median -3.3 , std 37.5, p90 68.1, halo 20). Void signal $\sim 10\text{--}30$. A $\delta\text{DM} < 50$ limit at 95 per cent needs $N_{\text{void}} \gtrsim 36$ (~ 200 localised total); unlocalised $z < 0.114$ needs ~ 900 void sightlines, have $\sim 30\text{--}50$.

5 SYSTEMATICS AND NEXT STEPS

Mask edge test done (5.6% cells edge; void fraction shift 3%, subdominant). Blinded SALT2 pilot (20 void + 20 wall z -matched Ia, Tripp $\alpha = 0.14$, $\beta = 3.1$, QC $|x_1| < 3$, $|c| < 0.3$, blind offset sealed) fetching ZTF DR lightcurves; fit validation 2/2 passed. Remaining: peculiar velocities, selection efficiency, host masses, TNS subtype purge, nulls (RA-shuffle within footprint, block bootstrap, 0.00/0.05 mag injection). No unblinding until budget closed.

ACKNOWLEDGEMENTS

H. D. is a Higher Secondary Certificate student. This work used public archives only and received no institutional funding. The author thanks the VAST, ZTF, CHIME/FRB, SDSS, and NASA-Sloan Atlas teams for making their data and software public. AI agents (Muse Spark 1.3) assisted in building analysis codes, LaTeX files, and figures; however, the scientific analysis, interpretations, and conclusions remain the sole responsibility of the author.

6 DATA AVAILABILITY

VAST <https://zenodo.org/records/7406035> (DOI <https://doi.org/10.5281/zenodo.7406035>, v1.3.0, zip sha256 9896002c...acef9); BTS <https://sites.astro.caltech.edu/ztf/bts/explorer.php> query quality+y purity+y (2026-09-30, CSV sha256 77668fe8...f38d65); CHIME Catalog 1 Vizier <https://vizier.cds.unistra.fr/viz-bin/VizieR-3?-source=J/ApJS/257/59/table2>; derived b3_match_v2.parquet (sha256 1a162c74...af77530) with scripts and randoms (seed 42/43/44) released with a Zenodo DOI on acceptance; repository public on acceptance.

Table 2. Masked sample composition. Wall means finite $1.0 < r/R_{\text{max}} < \infty$; Outside is $r/R_{\text{max}} = \infty$ (no void within $1.2 \max R_{\text{max}}$). Cosmo rows are spectroscopically classified SNe only (TDE/Ca-rich/other excluded); CC includes 6 SLSN; Ia subtype audit (91T-like etc.) pending TNS.

Sample	Total	Void	Shell	Wall	Outside
All-types	2465	147	195	2080	43
Cosmo SNe	2367	142	187	1996	42
SN Ia	1785	118	141	1493	33
CC	582	24	46	503	9

REFERENCES

- Planck Collaboration, 2020, A&A, 641, A6
Guy J. et al., 2007, A&A, 466, 11
Tripp R., 1998, A&A, 331, 815
Douglass K. A. et al., 2023, ApJS, 265, 7
Perley D. A. et al., 2020, ApJ, 904, 35
Fremling C. et al., 2019, arXiv:1910.12973
CHIME/FRB Collab., 2021, ApJS, 257, 59
CHIME/FRB Collab., 2026, arXiv:2601.09399
Wang et al., 2026, ApJ, 997, 121
Sharma et al., 2026, arXiv:2605.01994
Tsaprazi A.-M. et al., 2021, arXiv:2109.02651
Aubert et al., 2025, A&A, 694, A7
Walker et al., 2024, A&A, 683, A71
Macquart et al., 2020, Nature, 581, 391